

Paytm AI Hackathon: Winning Strategy

QR Rakshak Strategy

August 22, 2026

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Paytm AI Hackathon: Winning Strategy

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Paytm AI Hackathon: Winning Strategy

Hyderabad · 22 August 2026 · One-day sprint · Solo build

Recommended concept: **Paytm QR Rakshak** — the AI that detects missing payments.

Part A — Source analysis

WHAT THE OFFICIAL MATERIAL ACTUALLY SAYS

The Hyderabad theme PDF defines a one-day sprint with two problem spaces, and it is explicit that each theme “defines a problem space, not a solution.”

Track 1 — AI for Paytm Users. Intelligent features that improve speed, trust, and experience across the Paytm ecosystem. The document names paying, recharging, transferring, investing, and transacting. Users care about three things: that it is fast, that it feels safe, and that it is effortless. First-time and regional-language users are explicitly in scope.

Track 2 — AI for Small Businesses. AI tools that help India’s small and medium businesses grow, operate, and scale. The document names the kirana store, the chemist, the salon, the distributor, and the home-run brand. It spans understanding and reaching customers, managing money, inventory and billing, and planning ahead.

The binding constraint appears in both themes. Every solution must be built for the existing Paytm stack. It must plug into Paytm’s current products and ecosystem rather than stand alone as an unrelated idea. A standalone product is explicitly out of scope.

WHAT THE BRIEFING PHOTOS ADD

The judging slide lists five criteria: **Innovation** (“novel approach, are you thinking outside the box?”), **Feasibility** (“can it work in the real world? technical viability”), **Impact** (“how many people will it help, and how much?”), **Execution** (“quality of build, a working prototype, not a concept”), and **Relevance** (“does it address the core challenge of your track?”).

The slide carries a critical rule in its footer: **demos must be a working prototype. Slide decks or presentations alone are not allowed.**

RUBRIC CORRECTION

The official material does not state the 5x20 point labels supplied in the brief. The photographed criteria are Innovation, Feasibility, Impact, Execution, and Relevance. The 100-point model used throughout this document is a strategic mapping, not a quoted official score sheet. The mapping applied is Relevance to Paytm Integration and Execution to Demo Quality. This is an explicit assumption.

EVIDENCE HIERARCHY

Source	What it proves	What it does not prove
Official Hyderabad PDF	Tracks, one-day format, native Paytm requirement	Weights, API availability, team size
Hyderabad briefing photos	Criteria labels and the working-prototype rule	Exact point allocation
Mumbai submission CSV	44 prior concepts and saturation patterns	Winners, judge scores, or Hyderabad entrants
Current Paytm public pages	AI Soundbox, UPI Lite, UPI Circle, spend summary, merchant rails exist	Private API access at this event
Your input	Solo builder, no private API access	Exact remaining hours, preferred stack

DATA HYGIENE NOTE

The supplied CSV contains personal fields and at least one apparent API secret pasted by a prior entrant into a form field. This analysis deliberately excludes all names, emails, and secrets. Do not copy any credentials from that file.

PREVIOUS SUBMISSION ANALYSIS

You confirmed that none of the CSV entries belongs to your team. The file is therefore treated purely as Mumbai competitor intelligence. No “improve, pivot, combine, or abandon” decision applies to your own prior work, because there is none.

Part B — Judge scoring psychology

WHAT EACH SCORE BAND LOOKS LIKE

Criterion	5/20	10/20	15/20	20/20
Paytm Integration / Relevance	A logo, a payment link, or a standalone app	Uses a Paytm surface but could be any UPI app	Extends a real Paytm journey and uses plausible Paytm signals	Paytm has a structural advantage and can take a native action
AI Innovation / Innovation	LLM copy or a chatbot skin	Generic prompt plus retrieval	Purpose-built prediction or classification with grounding	New capability, visible intelligence, confidence, guardrails, no simpler substitute
User Impact / Impact	Vague convenience	Real pain, no quantified baseline	Frequent severe pain with defensible assumptions	Clear affected population, measurable money/time/trust outcome, validation path
Demo Quality / Execution	Slides or static screens	Clickable happy path	Working end-to-end flow with one clear AI reveal	Immediate problem, visible intelligence, Paytm action, quantified before/after, resilient live run
Build Feasibility / Feasibility	Architecture fiction	Mostly mocked with unclear boundaries	Real core intelligence plus an honest Paytm adapter	Small vertical slice, tested failure paths, seeded backup, no fragile dependency

HIDDEN IMPLICATIONS IN THE WORDING

“Problem space, not a solution” rewards sharp problem selection far more than broad feature coverage. A team that picks one overlooked problem outscores a team that builds six shallow features.

“Existing Paytm stack” means a Paytm payment link is not enough. The idea needs a native surface, a proprietary signal, and a native action. If your product would work identically on PhonePe, Relevance collapses.

The one-day sprint rewards deterministic demos, seeded fallbacks, and one complete vertical slice. It punishes anything that needs training data you do not have.

The working-prototype rule makes deck-first and architecture-only entries noncompetitive regardless of idea quality.

Two product facts matter enormously and are easy to miss. Paytm has launched an **AI Soundbox** with a built-in assistant that interacts with merchants in 11 local languages and provides business summaries and payment insights. Paytm also already ships a **Monthly AI Spend Summary** that auto-categorises consumer spending. Building another multilingual merchant voice copilot or another spend-analysis chatbot means presenting a weaker clone of Paytm’s own shipped product to Paytm’s own judges.

Part C — Opportunity map

THE PRIOR FIELD

The supplied CSV contains 44 submissions from the Mumbai event: 29 small-business concepts and 15 consumer concepts.

Mechanism	Approximate share of 44
Voice (STT/TTS/voice-first)	70%
Chatbot or copilot	61%
Inventory	30%
Finance, GST, or P&L	30%
OCR or vision	25%
WhatsApp as the channel	18%
Fraud overlay	9%

Percentages are hand-tagged from concept descriptions and are approximate.

Twelve teams built essentially the identical product: a vernacular merchant copilot answering “Aaj ka collection kitna hai?” over transaction data. The name collisions alone tell the story: Vyapar Saathi, Vyaparsaathi, Dukaan-Saathi, Paytm Sahayak, Paytm Saathi, Paytm Mitra.

DO NOT BUILD THIS

A multilingual merchant voice copilot. Voice appeared in roughly 70% of the prior field, copilots in 61%, and Paytm has since shipped the AI Soundbox assistant in 11 languages. You would be competing against both a crowd and the host’s own roadmap.

Also avoid: generic spend chatbots (Paytm ships Monthly AI Spend Summary), khata OCR digitisation, WhatsApp storefronts, broad “AI CFO” dashboards, screenshot-only fraud detection, and ungrounded revenue forecasts using Prophet on synthetic data.

Only one of the 44 prior submissions named any classical statistical or ML method. That is the opening.

TWENTY PROBLEM OPPORTUNITIES

Problem	User	Frequency	Severity	Paytm advantage	AI potential	Demo potential	Feasibility	Competition risk	Impact
Silent QR replacement	Micro-merchant	Daily	Critical	Very high	Very high	Excellent	High	Very low	Money trust
Payment acceptance outage	Merchant	Daily	High	Very high	High	Strong	High	Low	Lost sa
Stuck settlement ETA	Merchant	Weekly	High	Very high	High	Good	High	Low	Cash fl
Debited but not received	Consumer + merchant	Frequent	High	Very high	High	Strong	High	Medium	Trust + suppor
Recharge validity leakage	Prepaid user	Monthly	Medium	Very high	High	Good	Very high	Very low	Money saved
Wrong-payee / lookalike VPA	Consumer	Occasional	Critical	High	Very high	Excellent	Medium	Low	Fraud avoide
Autopay price creep	Consumer	Monthly	Medium	High	High	Good	High	Medium	Money saved
UPI Circle anomaly guard	Families	Weekly	High	High	High	Strong	High	Medium	Trust
Soundbox fleet fault prediction	Multi-store merchant	Daily	High	Very high	High	Strong	Medium	Very low	Uptime
Merchant dispute evidence pack	Merchant	Weekly	High	Very high	Medium	Strong	High	Low	Time saved
Queue/staffing from cadence	Retail/food merchant	Daily	Medium	High	High	Good	High	Very low	Time + sales
Privacy-safe peer benchmark	Merchant	Weekly	Medium	Very high	High	Good	Medium	Very low	Growth
Refund delay prediction	Consumer	Occasional	High	Very high	High	Good	High	Low	Trust
Agent-payment consent control	Power user	Occasional	High	Very high	Very high	Excellent	Medium	Medium	Future platform
Micro-contract escrow	Informal worker	Weekly	Critical	High	High	Excellent	Medium	Medium	Income securit
Utility bill anomaly	Household	Monthly	Medium	Very high	High	Strong	High	Low	Money saved

Problem	User	Frequency	Severity	Paytm advantage	AI potential	Demo potential	Feasibility	Competition risk	Impact
Local demand spike detection	Kirana/pharmacy	Weekly	High	Very high	High	Strong	Medium	Low	Revenue
Merchant payout safety	Merchant	Weekly	High	Very high	High	Strong	Medium	Medium	Error prevention
Informal worker reputation	Gig worker	Monthly	Critical	High	Medium	Strong	Medium	Low	Credit access
Connectivity-aware rail choice	Commuter	Weekly	Medium	Very high	High	Strong	High	Medium	Reliability

WHITE-SPACE CLASSIFICATION

Obvious ideas (many teams will build these): merchant voice copilot, spend chatbot, inventory forecast, fraud warning banner, bill OCR dashboard.

Interesting ideas (differentiated): recharge optimiser, autopay leakage audit, queue inference from payment cadence, refund ETA, adaptive UPI Circle guard.

White-space ideas (real competitive advantage): silent QR replacement detection, acceptance-interruption detection, rail-state resolution prediction, wrong-payee intent mismatch, privacy-safe network benchmarking.

Moonshots (transformational, too hard for one day): mule-ring graph intelligence, agent-payment consent rail, cross-PSP trust receipt standard, informal-worker reputation, counterfeit detection at checkout.

The sweet spot sits where Paytm has a structural advantage, AI creates a genuinely new capability, the problem recurs daily, existing solutions are reactive, the demo can be physical, and a solo developer can finish it.

Part D — Twenty-two concepts, brutally scored

Scores are deliberately compressed. A high score requires a plausible solo build, not merely a compelling vision.

Idea	Paytm /20	AI /20	Impact /20	Demo /20	Feasibility /20	Total /100	WOW /100
QR Rakshak	19	17	18	19	18	91	94
Paisa Kahan Hai	19	17	19	16	18	89	84
Sahi Payee	18	17	18	18	18	89	85
Sahi Plan	19	16	17	16	20	88	80
Bill Shock	19	17	17	18	17	88	83
Circle Guard	19	17	17	17	18	88	82
Soundbox Pulse	20	17	16	17	17	87	85
No-Network Checkout	19	16	17	18	17	87	84
Refund Radar	19	16	17	16	19	87	76
Dispute Pack	19	15	18	17	20	89	78
Mandate Mirror	18	16	17	16	19	86	77
Trust Receipt	20	12	18	17	20	87	76
Cashflow Circuit	18	16	17	17	17	85	79
Bheed	17	18	15	17	18	85	80
Payee Trust Graph	18	19	19	19	13	88	95
Merchant Benchmark	20	17	16	15	14	82	78
Sauda Signal	19	18	17	18	13	85	88
Payout Pilot	19	18	17	18	14	86	86
Festival Float	18	18	17	18	12	83	85
Zubaan Reputation	18	17	18	17	12	82	84
Pocket Pace	20	15	14	14	20	83	70
Payment Memory	18	14	14	16	20	82	72
Pakka Payment	15	13	16	19	16	79	88
Counterfeit Counter	16	18	17	18	8	77	90

THE TOP EIGHT IN DETAIL

1. QR Rakshak — 91/100

One-line pitch. Detect when a merchant's QR silently stops earning, diagnose a replaced QR, and restore acceptance before the day's revenue is lost.

Track. Track 2, AI for Small Businesses.

User. Paytm QR and Soundbox micro-merchants during busy hours. The owner-operator who cannot watch a dashboard.

Pain. A pasted-over or swapped QR redirects payments to someone else. The Soundbox stays silent. The merchant assumes business is slow and may only discover the loss at closing time.

Current workaround. Wait for a customer complaint, manually inspect the QR sticker, or reconcile at day-end.

AI mechanism. Contextual transaction-cadence anomaly detection. The model estimates the merchant's expected payment count for this time bucket, detects abnormal silence against that expectation, and ranks likely causes. Deterministic VPA verification then confirms or rejects the replacement hypothesis.

Paytm integration. Paytm for Business home alert, AI Soundbox notification, QR self-check, and reissue or support action.

Data and signals. Transaction timestamps and amounts, business hours, merchant-specific baseline, anonymised cohort activity, Soundbox heartbeat, registered VPA, scanned QR payload.

User flow. The model observes abnormal silence, the merchant receives a cautious alert, scans the QR displayed at the counter, the app compares the parsed VPA against Paytm's registered VPA, one tap starts recovery, and the next payment confirms restoration.

Magic moment. The model draws eight expected payments that never arrived, then the camera scan exposes the pasted QR's wrong VPA.

Why now. QR acceptance is ubiquitous and Paytm now has an AI Soundbox surface capable of proactive merchant alerts.

Why Paytm. Only Paytm can combine the merchant's expected payment rhythm, the registered VPA, device telemetry, and the recovery workflow.

Competitive differentiation. It models the negative space, the payments that should have happened, rather than adding another assistant on top of payments that did.

MVP. Seeded live stream, expected-count anomaly scorer, explainable alert, camera and upload QR parser, VPA mismatch check, simulated recovery.

Stretch. Cohort baselines, device-fault classifier, geofenced self-test, automatic replacement order.

Demo. Swap a real printed QR with a wrong one. The stream goes silent, the alert fires, the scan reveals the mismatch, and the corrected QR receives a payment.

Risks. False alarms during genuine quiet periods, insufficient history for low-volume merchants, and unsupported assumptions about telemetry and API access.

Hackathon feasibility. High.

2. Paisa Kahan Hai — 89/100

Pitch. Give a precise, confidence-scored ETA and next action for pending settlements and debited-but-not-received payments.

User. Consumers and merchants waiting for money. **Pain.** Uncertainty is worse than delay; users repeatedly refresh history or contact support. **AI.** A survival model predicts resolution time from bank, rail, status, hour, and amount. **Paytm fit.** Transaction and settlement detail screens; Paytm can update status, raise the correct dispute, and notify both parties. **Magic.** A vague PENDING becomes "94% likely by 6:42 PM" with a live timeline. **Risk.** Needs historical labels, and an inaccurate ETA damages trust.

Feasibility. High.

3. Sahi Payee — 89/100

Pitch. Catch wrong-recipient transfers using phonetic, visual, and behavioural mismatch before PIN entry.

User. Anyone paying a new or similarly named contact. **Pain.** One digit or one homoglyph sends irreversible money to a stranger.

AI. Name and VPA similarity, contact context, prior-payee graph, amount novelty, QR-location consistency. **Paytm fit.** The payee confirmation screen. **Magic.** The app catches `ramesh@upi` versus `rarnesh@upi` and explains the visual trick. **Risk.** Privacy and false friction. **Feasibility.** High.

4. Dispute Pack — 89/100

Pitch. Turn a failed-payment timeline into the exact evidence bundle and dispute route in one tap.

AI. State-sequence classification selects dispute type, missing evidence, and likely resolver. **Magic.** A twelve-screen complaint becomes one verified timeline with a case ID. **Weakness.** May read as support automation rather than innovative AI. **Feasibility.** Very high.

5. Sahi Plan — 88/100

Pitch. Stop prepaid users wasting validity by building the lowest-cost annual recharge ladder from their real recharge rhythm.

AI. Usage-profile inference plus constrained optimisation over the plan catalogue. **Magic.** A family's habitual spend collapses into a cheaper plan with dozens fewer wasted validity days. **Weakness.** May be scored as optimisation rather than breakthrough AI, and the demo is a comparison table. **Feasibility.** Very high. This is the safest solo build on the list.

6. Payee Trust Graph — 88/100

Pitch. Warn before money moves by detecting mule-ring topology and lookalike payees, not just scam words.

AI. Graph anomaly features, fan-in/fan-out ratios, fund velocity, cycle detection, community detection, phonetic similarity. **Magic.** A force-directed graph lights up the recipient's hidden mule ring before PIN entry. This has the highest raw visual wow on the entire list. **Fatal weakness for this event.** A solo developer would spend the day fabricating synthetic graph data and a visualisation layer with no time left to make the detection credible, and the data sensitivity invites regulatory objections. Keep as a moonshot only.

7. Soundbox Pulse — 87/100

Pitch. Predict Soundbox or QR acceptance failures before merchants lose a full shift.

AI. Multivariate anomaly detection over heartbeat, signal, battery, payment cadence, and peer incidents. **Strength.** The most Paytm-native concept on the list; only Paytm controls device telemetry and field service. **Weakness.** Emotionally weaker than a visible physical attack, and more dependent on telemetry you do not have. **Feasibility.** High.

8. Bill Shock — 88/100

Pitch. Detect abnormal utility bills before payment and explain the likely driver using the user's own history.

AI. Seasonal anomaly detection plus bill-line OCR and driver attribution. **Magic.** A 3x bill is traced to an impossible meter jump before checkout. **Paytm fit.** The bill-payment review screen, where Paytm already owns repeated bill history. **Feasibility.** High.

THE REMAINING CONCEPTS IN BRIEF

Circle Guard keeps UPI Circle convenient while flagging delegated spending that breaks a family member's normal pattern; adaptive trust rather than another family budget dashboard. **Mandate Mirror** exposes subscription price creep, duplicate mandates, and zombie AutoPay before the next debit using periodicity and changepoint detection. **Refund Radar** predicts refund arrival and detects when a refund has silently left its normal path. **No-Network Checkout** predicts connectivity risk before a low-value payment and prepares the most reliable Paytm rail; carries prior-art risk against a Mumbai entry. **Trust Receipt** creates a shared verifiable post-payment receipt that ends the "I paid / I didn't receive it" argument; excellent Paytm fit but the core value is deterministic, so AI innovation scores low. **Cashflow Circuit** warns merchants before settlement timing and scheduled payouts create a cash shortfall. **Bheed** infers queue pressure from payment inter-arrival times and applies queueing theory to recommend staffing, with no cameras and no new hardware. **Merchant Benchmark** shows a merchant how they compare with similar nearby businesses using k-anonymity and differential privacy, where the privacy engineering is itself the story. **Sauda Signal** detects a neighbourhood demand spike from privacy-safe payment patterns and triggers a small timely stock order. **Payout Pilot** lets merchants prepare supplier or staff payouts by voice while a deterministic policy engine prevents wrong or excessive transfers; the safety architecture is the product. **Festival Float** recommends one evidence-backed inventory bet and matching working capital before a local festival. **Zubaan Reputation** converts a completed informal-work payment trail into a portable consented reputation proof. **Pocket Pace** makes UPI Lite auto top-up adaptive to real micro-spending rhythm. **Payment Memory** turns ambiguous payment search into a visual memory query, but competes directly with Paytm's shipped AI spend features. **Pakka Payment** checks a customer's payment screenshot against the merchant ledger and highlights image tampering; the visual demo is strong but the AI is unnecessary because the ledger and Soundbox already provide the authoritative answer, and accusing a customer of forgery on a noisy forensic signal is a product and legal mistake. **Counterfeit Counter** verifies high-risk goods at payment time; spectacular but requires datasets and liability handling far beyond one day.

Part E — Competitive analysis

Assume ten other teams. These are the predictable builds.

Likely team	Why attractive	Why judges may like it	How QR Rakshak beats it	What to avoid
Merchant voice copilot	Easy Sarvam demo	Inclusive and familiar	Paytm has productised this; Rakshak solves a new trust failure	Do not add chat
AI spend coach	Easy transaction dataset	Broad consumer reach	Spend summary already ships; Rakshak has a physical wow moment	Do not claim generic savings
Fraud risk score	Emotional problem	Trust fit	Rakshak has a specific attack and deterministic confirmation	Avoid a vague 0-100 score
Inventory forecaster	Merchant impact story	Growth narrative	Rakshak needs no item-level inventory	Avoid synthetic Prophet charts
Khata OCR	Camera demo	Regional relevance	Crowded cluster; Rakshak uses the camera to confirm a real Paytm object	Avoid document management scope
WhatsApp shop	Full commerce flow	Visually broad	Weak native-stack fit; Rakshak lives inside Paytm for Business	Avoid external channel dependency
Voice payments	Fast accessible demo	Consumer convenience	Crowded and security-sensitive; Rakshak protects existing payments	Avoid autonomous money movement
Loan readiness score	Large merchant impact	Paytm lending fit	Needs underwriting labels; Rakshak needs only cadence	Avoid unsupported eligibility claims
Agentic checkout	Future-facing	High novelty	Harder to build safely; Rakshak completes in a day	Avoid MCP theater
Bill scanner	Reliable OCR demo	Easy before/after	Limited Paytm moat; Rakshak depends on Paytm's registered identity	Avoid standalone utility

IF ANOTHER TEAM COPIES US TOMORROW

The concept name is copyable. The hard-to-copy unit is the combined mechanism: merchant-specific absence modelling, plus Paytm registered-VPA verification, plus Soundbox and device context, plus a native recovery action. Ship all four. If you ship only the alert, the moat is nothing.

Part F & G — Red team on the top five

Idea	Strongest objection	How to eliminate it
QR Rakshak	"Traffic fluctuates naturally. This will annoy merchants every slow afternoon."	Use cautious "acceptance interruption" language, require multiple concurring signals, add snooze and feedback, and make the QR mismatch check deterministic rather than probabilistic.
QR Rakshak	"Why is this AI? A threshold would do."	A threshold ignores time of day, merchant rhythm, variance, device state, and comparable local activity. Show the expected distribution, the confidence band, and a counterfactual.
Paisa Kahan Hai	"An incorrect ETA damages trust more than no ETA."	Use calibrated ranges with explicit confidence and automatic escalation when the actual path deviates from prediction.
Sahi Plan	"This is a comparison engine, not AI."	Demonstrate constraint inference, annual optimisation under multiple constraints, full explainability, and real checkout.
Payee Trust Graph	"The data and the regulation make this impossible in a hackathon."	Use clearly labelled synthetic graph data and build only the scoring and visualisation. Do not make production claims. Retain as moonshot.
Soundbox Pulse	"This is internal device operations, not merchant value."	Lead with lost sales avoided and surface the diagnosis on the Soundbox itself.

THE SEVEN JUDGE PERSONAS, ANSWERED FOR QR RAKSHAK

"What is wrong with this?" The base rate of QR replacement is unmeasured, and low-volume merchants will have weak baselines. Both are stated openly rather than hidden.

"Why isn't this just another AI wrapper?" There is no LLM in the must-ship path. The intelligence is an expected-count model over the merchant's own cadence, and the confirmation is a deterministic identity comparison.

"Where is Paytm integration?" The signal, the expected identity, the device context, and the recovery action are all Paytm-native. A standalone app has none of the four.

"Can they actually build this?" The real core is a typed event replay, a lightweight time-series scorer, a QR decoder, and a VPA comparison. Everything Paytm-side sits behind an honest mock adapter.

"How many users does this help?" The official material gives no verified merchant count, so we state the target population as Paytm QR and Soundbox merchants and refuse to invent a number.

"Another team built something similar." No prior submission modelled payment absence. If challenged, point to the four-part mechanism rather than the alert.

"The demo is impressive but is the product useful?" The outcome is a restored acceptance point and a measured detection delay, not a dashboard.

Part H — The winner

PAYTM QR RAKSHAK

Paytm already confirms the money that arrives. QR Rakshak detects the payments that should have arrived, and helps the merchant restore acceptance.

Why this one. It has exactly one urgent problem: a merchant's acceptance point silently stops earning. One AI insight: detect absence relative to merchant-specific expected cadence. One Paytm-native confirmation: compare the displayed QR against the merchant's registered VPA. One Paytm action: restore or reissue the QR and verify recovery with the next payment. One memorable moment: a physical covered QR exposed on camera after the model detects the silence.

Why not the others. Paisa Kahan Hai needs labelled rail-state history and has a less visual demo. Sahi Plan is highly buildable but may be scored as optimisation. Payee Trust Graph has spectacular visuals but unacceptable data and feasibility risk for one solo day. Soundbox Pulse is the most Paytm-native but is emotionally weaker and more dependent on telemetry you cannot access.

Expected judging score. 91/100. This is an optimisation target, not a promise.

Biggest competitive advantage. The four-part mechanism. Absence modelling alone is copyable; absence modelling plus registered-identity verification plus device context plus native recovery is not reproducible overnight.

Biggest risk. Natural quiet periods look like incidents. A false alarm at every slow hour destroys merchant trust and the demo's credibility.

Risk mitigation. Require a high-confidence combination of cadence residual, comparable local activity, and device heartbeat. Say "acceptance interruption," never "fraud." Always ask for the deterministic QR self-check before asserting anything.

Why Paytm should actually want this. It protects merchant GMV, QR trust, Soundbox value, merchant retention, and support cost using assets Paytm already owns.

Why judges will remember it. "The AI that notices missing payments" is repeatable in one sentence, and the pasted-QR reveal is physical rather than another dashboard.

Backup #1. Paisa Kahan Hai — the best trust and support product if transaction-status data is available.

Backup #2. Sahi Plan — the safest solo build, with lower wow but exceptional feasibility.

CONCEPT COMBINATIONS CONSIDERED

Combination	Strategic gain	Decision
Cadence anomaly + QR identity + Soundbox heartbeat + recovery	Turns suspicion into confirmation and action	Winner. Smallest complete strategic unit
Screenshot forensics + ledger match	Visual heatmap	Reject. The ledger and Soundbox already answer this; image AI is unnecessary
Settlement ETA + dispute pack	Prediction becomes action	Strong Backup #1 expansion
Recharge optimiser + adaptive UPI Lite	Consumer savings suite	Reject. Two unrelated rails dilute the story
Payee graph + UPI Circle guard	Family fraud defence	Reject for a solo day. Sensitive graph data

CONTINUOUS OPTIMISATION PASS

How could this be 10x more impressive without being 10x harder? Use a real printed replacement QR and a live camera scan while keeping the entire backend seeded.

What single feature makes judges remember this tomorrow? The expected-payments ghost line: draw the payments that should exist but do not.

What can we remove while making the product stronger? Remove chat, dashboards, multi-merchant fleet views, fraud heatmaps, and predictive inventory.

What makes it feel like a real Paytm product? The Paytm for Business entry point, the Soundbox heartbeat, the registered VPA, the recovery adapter, and restrained fintech microcopy.

Part I — Product specification

Product name. Paytm QR Rakshak

One-line pitch. Detect when a merchant's QR silently stops earning and restore Paytm acceptance before revenue disappears.

Track. Track 2, AI for Small Businesses.

Product thesis. Payment products are optimised for events that occur: initiated, pending, credited, failed. A replaced QR creates no event at all in the rightful merchant's system. QR Rakshak treats that absence as the signal. It learns the merchant's normal payment rhythm, compares observed against expected activity alongside device and context signals, and asks for a deterministic QR identity check only when confidence is high. The intelligence detects; Paytm verifies and acts.

User persona. Ramesh, owner-operator of a busy kirana with one QR and one Soundbox. He has no staff to watch a dashboard and no time to reconcile mid-shift.

Core problem, emotionally and economically. The merchant hears nothing, sees customers leave, and concludes business is slow. Every payment sent to a covered QR is immediate loss and a trust incident. The existing workaround is entirely reactive: wait for a complaint, search history, or reconcile at closing. The impact figure in the demo must be labelled as a model estimate: expected payment count multiplied by normal ticket size across the interruption window.

Product promise. If you use QR Rakshak, Paytm will warn you when your acceptance point behaves abnormally, help you verify the QR on your counter, and guide recovery before a quiet counter becomes a lost day.

CORE USER JOURNEY

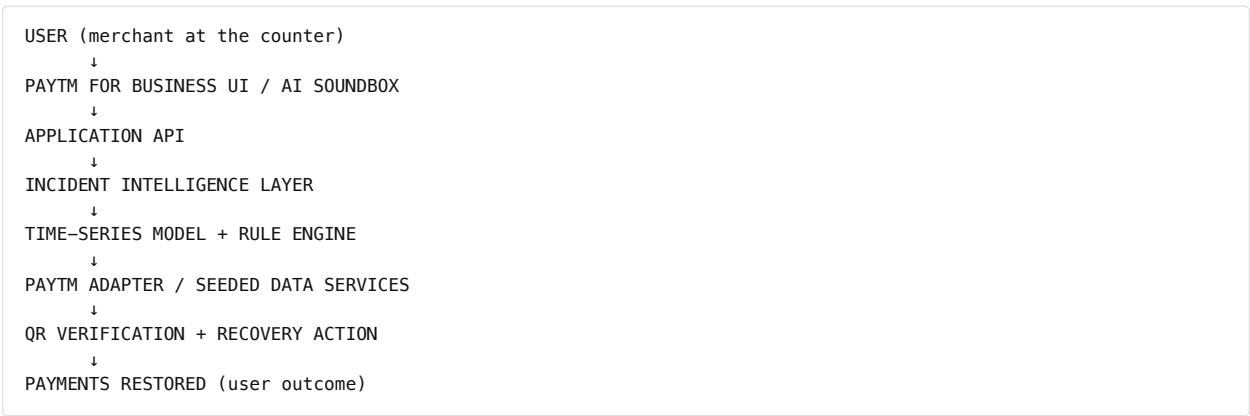
Step	What happens
1. Entry point	Paytm for Business home, plus AI Soundbox notification
2. Trigger	Observed credits fall materially below merchant-specific expected cadence
3. User action	Merchant opens "Payments unusually quiet"
4. AI processing	Model scores cadence residual, device health, local context, and confidence
5. AI insight	"Possible acceptance interruption. QR check recommended," with ranked reasons
6. User action	Merchant scans the QR displayed at the counter
7. Paytm action	Registered VPA mismatch confirmed; recovery and reissue workflow starts
8. Outcome	Test payment lands, Soundbox confirms, monitoring returns to normal

SCREENS

Screen	Purpose	Components and microcopy	Primary CTA	Data shown	AI interaction	Next state
1. Paytm for Business home	Make abnormal silence impossible to miss without causing panic	Status strip “Payments unusually quiet”; expected-versus-actual mini line; Soundbox online indicator	Review acceptance	Observed 0 versus expected 8; 92% confidence; estimated sales at risk	No chat. Tap the explanation or scan the QR	Opens incident detail
2. Acceptance incident	Explain why the system interrupted the merchant	Timeline, contributing signals, benign-cause options, confidence, QR self-check entry	Check my QR	Last credit, normal cadence, nearby cohort activity, device heartbeat	Counterfactual: “if area traffic were also down, risk would be 31%, not 92%”	Launches camera or upload
3. QR self-check	Turn probabilistic suspicion into deterministic confirmation	Camera frame, parsed payee, expected Paytm VPA, mismatch banner	Restore Paytm QR	Expected versus scanned VPA and merchant name	Vision only locates and reads the QR; validation is deterministic	Starts simulated reissue
4. Recovery	Complete the Paytm action	Fresh dynamic QR, reprint option, support case, test-payment button	Run a test payment	New QR ID, case status, device status	AI monitors the recovery window	Moves to confirmation
5. Restored	Show a measurable outcome	Soundbox confirmation, green payment event, downtime and modelled loss avoided	Done	Detected in 11 minutes; revenue at risk labelled as an estimate	Feedback improves future thresholds	Returns home

Design quality bar. Mobile-first and Paytm-like. One primary action per screen. Restrained red used only for a confirmed mismatch. No generic AI badge, no chatbot, no decorative dashboard cards. Always visually separate observed facts, model estimates, and simulated Paytm actions.

Part J — Technical architecture



STACK

Frontend. Vite or Next.js with React and TypeScript. A 390px mobile shell. A seed replay controller, an incident state machine, an SVG expected-versus-actual chart, and camera capture with an upload fallback.

Backend. A thin typed API or in-process demo service. The incident scorer, a `PaytmAdapter` interface, a QR parser and VPA validator, and an append-only event log. No authentication fiction in the MVP.

AI. An expected-count baseline with a calibrated anomaly score, plus a deterministic rule layer. No external model call is required in the must-ship path.

Infrastructure. Local-first seeded mode, optional Vercel deployment, structured logs, no secrets in the client, a frozen seed, and a fallback video.

DATA MODEL

Entity	Essential fields	Purpose
Merchant	id, displayName, assignedVpa, timezone, businessHours, category	Identity and expected operating context
PaymentEvent	id, merchantId, amount, status, occurredAt, channel	Observed credit cadence
DeviceHeartbeat	deviceId, online, signal, battery, seenAt	Separate device failure from missing payments
ExpectedCadence	bucket, lambda, variance, source, updatedAt	Merchant-specific expected count
AcceptanceIncident	openedAt, score, confidence, reasons, state	Explainable model output
QrIdentity	rawPayload, parsedVpa, expectedVpa, matched	Deterministic confirmation
RecoveryAction	type, simulated, status, createdAt	Honest action boundary
ModelFeedback	incidentId, useful, benignReason	Threshold improvement and human override

APIS

Endpoint or interface	Input	Output	MVP status
<code>GET /demo/stream</code>	scenario, replay time	payment and heartbeat events	Real seeded service
<code>POST /incidents/score</code>	recent events, baseline	score, confidence, reasons, expected count	Real
<code>POST /qr/verify</code>	QR image or decoded payload	parsed VPA, expected VPA, match	Real parser, seeded expected VPA
<code>POST /recovery/restore</code>	merchant, incident	action ID, status	Mock Paytm adapter
<code>POST /payments/test</code>	merchant, amount	seeded successful event	Simulated Paytm
<code>PaytmAdapter</code>	<code>getTransactions</code> , <code>getDevice</code> , <code>restoreQr</code>	typed domain objects	Mock now; production adapter documented

THE AI SYSTEM, DESIGNED PROPERLY

Layer	Design
Inputs	Recent credits, historical five-minute buckets, business hours, weekday, device heartbeat, optional cohort-activity index, assigned and scanned VPA
Features	Count residual, inter-arrival z-score, EWMA drop, CUSUM statistic, amount-weighted shortfall, device-online-but-no-credit interaction, cohort residual
Model	Negative-binomial or Poisson expected-count baseline with a calibrated anomaly score. Simple enough to explain out loud and run locally
LLM usage	None in the must-ship path. Optionally, a vernacular paraphrase of an already-computed result. The LLM never scores risk and never authorises an action
Rules	Never alert outside business hours. Minimum history and volume required. Hard VPA comparison. Alert cooldown. Suppress when an area or device outage already explains the drop
Retrieval	Not needed for the MVP. Production could retrieve verified Paytm support procedures, never open-web content
Classification	Cause ranking across replaced QR, device outage, and normal quiet period
Prediction	Expected payment count and confidence band per time bucket
Recommendation	The single next action: check the QR
Agent behaviour	None. No autonomous action, no money movement
Confidence scoring	Probability band, observed and expected counts, and a data-sufficiency label. No fake precision when history is sparse
Human override	Merchant can mark normal quiet period, shop closed, device issue, or useful alert
Guardrails	Say “possible interruption,” never “fraud.” No automatic payment blocking. Never accuse the owner of the scanned VPA
Explainability	Three ranked factors plus one counterfactual. Clearly distinguish observed, estimated, and confirmed
Evaluation	Scenario-level precision and recall, detection delay, false alerts per merchant-day, and calibration. Not generic model accuracy

PAYTM INTEGRATION, IN DETAIL

Where inside Paytm does this appear? Paytm for Business home, the device and QR section, and the incident detail screen. Notifications can also surface through the AI Soundbox.

What existing capability does it extend? QR acceptance, Soundbox payment confirmation, device health, transaction analytics, merchant support, and QR replacement or reissue.

What Paytm data could power it? The merchant transaction stream, the registered VPA, device heartbeat, the business profile, and a privacy-safe cohort activity signal.

What action can Paytm perform after the AI recommends? A QR self-check, a dynamic QR replacement, a reprint or support ticket, a test payment, and continued incident monitoring.

What makes it native? Detection depends on Paytm’s payment and device context. Confirmation depends on Paytm’s registered acceptance identity. Recovery depends on Paytm’s own QR lifecycle.

What happens if this were a standalone app? It would see neither authoritative credits nor the assigned VPA nor device state, and it could not restore Paytm acceptance. It would be reduced to asking the merchant to type in their own sales, which defeats the entire premise. The standalone version is not merely weaker; it is impossible.

Part K — Real versus mock

Boundary	What belongs there
REAL	Replay engine, anomaly scoring, confidence reasons, QR decoding, VPA comparison, UI state machine, error and loading paths
MOCKED	Paytm authentication, production transaction API, Soundbox telemetry, QR rotation and reissue, support ticket creation
SEEDED	Merchant profile, normal transaction history, incident stream, assigned VPA, replacement QR, cohort activity
SIMULATED PAYTM	<code>PaytmAdapter</code> responses, test payment, device heartbeat, recovery status

Every mocked action must carry a visible “Demo Paytm service” label or appear in an architecture disclosure. Never imply production API access. Honest boundaries score higher than a convincing lie, and a judge who catches an overstated integration will discount the entire submission.

Part L — The three-minute demo

Time	Screen	Presenter action	What to say	Proof delivered
0:00-0:20	Physical counter and merchant home	Point to a Paytm QR covered by another QR	“Ramesh’s Soundbox usually speaks every 90 seconds at lunch. It has been silent for 18 minutes. He thinks the shop is slow. His QR was replaced.”	Problem understood in seconds
0:20-0:50	Live transaction stream	Replay normal noon cadence, then zero credits while the heartbeat stays online	“Paytm sees what Ramesh cannot. Not a failed payment, but the payments that should have happened.”	Visible expected-versus-actual gap
0:50-1:30	AI incident screen	Trigger the anomaly score, expand three contributing signals	“A merchant-specific model combines cadence, today’s baseline, nearby activity, and device health. It reports an acceptance interruption at 92% confidence. Not fraud.”	Core AI is visible and explainable
1:30-2:10	QR camera check and recovery	Scan the covering QR, reveal the wrong VPA, tap Restore, show the replacement QR	“Prediction tells us to look. Paytm’s registered VPA gives the authoritative answer. One tap starts recovery inside Paytm for Business.”	Strong Paytm-native action
2:10-2:40	Test payment and Soundbox confirmation	Run a seeded payment; the stream resumes and the incident closes	“The next payment lands, the Soundbox confirms it, and Rakshak verifies recovery. Revenue at risk is transparently modelled, not claimed as observed.”	Before, intelligence, after
2:40-3:00	Scale view	Show three cause tiles: replaced QR, device outage, normal quiet period	“Paytm already confirms money that arrives. QR Rakshak understands dangerous silence. At scale, every Paytm acceptance point becomes self-monitoring.”	Memorable one-line vision

THE WINNING STORY

Problem. Today, a Paytm merchant can hear every payment that arrives, but a replaced QR creates only silence.

Insight. We realised the missing payments are themselves a signal, once Paytm understands the merchant’s normal rhythm.

Solution. So we built QR Rakshak, which detects an acceptance interruption, verifies the displayed QR, and restores Paytm acceptance.

AI. Unlike a static threshold, our model learns expected cadence and explains why today’s silence is abnormal.

Paytm. Because this is built into Paytm, a prediction becomes an authoritative identity check and a real recovery action.

Impact. This can reduce detection from end-of-day to minutes and protect revenue that would otherwise be redirected or lost. Pilot validation is required before any hard number.

Vision. At scale, every Paytm acceptance point becomes self-monitoring.

PRESENTER SAFEGUARDS

Start from the covered physical QR, not a title screen. Never say the model proves fraud; it detects an acceptance interruption and the VPA comparison confirms a mismatch. Label revenue protected as a modelled estimate. If the camera fails, switch to upload without explaining. If deployment fails, run local seeded mode, because nothing in the core demo should require a network. Finish by 2:50 to leave ten seconds for the closing line.

Part M — Build plan for a solo developer

Phase	Exact tasks	Definition of done	Cut if behind
Phase 1, 30 min	Lock one story. Seed 60-90 minutes of normal and incident transaction events. Define expected and scanned VPAs	Dataset replays deterministically and demo narration is frozen	Cut all extra incident types
Phase 2, 1 hour	Build the typed event model, expected-count baseline, EWMA/CUSUM score, confidence explanation, and unit cases	Normal period stays quiet; the replacement incident triggers at a known timestamp	Use fixed calibrated parameters instead of any training
Phase 3, 2 hours	Build the mobile Paytm Business shell, home alert, incident timeline, and expected-versus-actual visualisation	The AI moment is clickable and legible on one screen	Remove settings and navigation
Phase 4, 4 hours	Add QR camera and upload, parse the UPI payload, compare VPA, recovery action, and the test-payment event	The complete vertical slice works offline with seeded data	Use image upload if the live camera is unstable
Phase 5, 6 hours	Add the Soundbox audio cue, the normal and false-alarm path, error and loading states, and transparent mock badges	Every judge question has an on-screen answer	Cut audio first, then the cohort signal
Phase 6, polish	Rehearse three times, record the fallback video, test the mobile viewport, prewarm the app, freeze the seed, prepare backup screenshots	A three-minute run finishes under 2:50 with no network dependency	No new features

MUST SHIP

Normal-to-incident transaction replay. A real expected-versus-observed scorer. Three explainable factors. Physical or uploaded QR decode. Registered-VPA mismatch. Simulated recovery and a successful test payment. Visible mock and seed disclosure. A three-minute fallback recording.

DO NOT BUILD

A chatbot or voice assistant. Inventory, GST, loans, loyalty, or generic analytics. Production Paytm authentication. A trained deep model with no data. A multi-merchant admin dashboard. Automatic accusations or payment blocking. Any backend work before the judge journey is demonstrable. Any external dependency inside the core demo.

TEAM STRUCTURE

The solo constraint changes sequencing, not standards. Work in vertical slices: seed, score, show, verify, recover. Product story and implementation cannot be separate workstreams, so freeze the narrative in the first 30 minutes and make every subsequent code decision serve that exact click path.

FAILURE MODES THAT WOULD DESTROY THE SCORE

Failure mode	Why fatal	Mitigation
Looks like a static threshold	AI innovation collapses	Show the merchant-specific expected distribution, confidence, and a counterfactual
Claims fraud from silence	Trust and legal credibility collapse	Use acceptance-interruption language; confirm only the VPA mismatch
No real QR decode	The demo feels staged	Decode a printed QR live, with upload fallback
Paytm is only branding	Relevance collapses	Use the registered VPA, Soundbox heartbeat, and recovery adapter
False precision	Judges challenge the evidence	Label seeded data, confidence bands, and estimated revenue
Camera or network failure	Execution collapses	Local mode, preloaded QR, fallback video
Too many causes or features	The story becomes unclear	Demo only the replaced QR; mention device outage as future work
Generic dashboard UI	The prototype looks copied	One incident flow, one CTA, restrained hierarchy
No normal case shown	The model looks trigger-happy	Show quiet-period suppression in Q&A or on a backup screen
Hidden mock integration	Credibility loss	Explicit adapter boundary and visible demo labels

Part N — Judging scorecard

Criterion	Target	Why	Evidence in demo
Paytm Integration / Relevance	19/20	Uses transaction cadence, registered VPA, Soundbox and device context, and a recovery action	Native alert, QR identity check, Paytm recovery
AI Innovation / Innovation	17/20	Detects missing events with contextual time-series intelligence and explainability	Expected ghost line, confidence, factors, counterfactual
User Impact / Impact	18/20	Protects merchant revenue and trust on a frequent acceptance surface	Modelled sales at risk and detection delay
Demo Quality / Execution	19/20	Physical attack, visible AI, deterministic confirmation, successful recovery	Printed QR swap through to the next credited payment
Build Feasibility / Feasibility	18/20	Real core, seeded Paytm adapter, no fragile model API	A complete local vertical slice
TOTAL	91/100	High-probability strategy, not a promise	All five criteria visible within three minutes

WHAT ACTUALLY PREVENTS 100/100

There is no production Paytm telemetry or API access. There is no merchant pilot and therefore no measured base rate for QR replacement. Model calibration is uncertain for low-volume merchants. Recovery and reissue are simulated rather than real. And the AI component, while rigorous, is less fashionable than generative AI, so its novelty must be demonstrated rather than asserted.

REALISTIC REDESIGNS THAT CLOSE THE GAP

Add a normal-quiet replay and a false-alarm feedback path to prove restraint. Show detection delay and false alerts per merchant-day across synthetic scenario tests. Use the physical QR and a real decoder so the confirmation step is genuinely not mocked. Expose the `PaytmAdapter` interface and name the production data it would require. If you use a vernacular Soundbox alert at all, generate it only after the score is computed, so language generation never becomes the product.

WHAT TO SAY IF JUDGES CHALLENGE US

"Why not just use the Soundbox?" The Soundbox confirms credits. A replaced QR produces no credit to confirm. Rakshak interprets abnormal silence and then asks for an authoritative check.

"Isn't this just a threshold?" A threshold ignores time of day, merchant rhythm, variance, device state, and comparable local activity. The demo shows a contextual expected distribution and a counterfactual.

"How do you know it is fraud?" We do not, and we never claim to. The model identifies an acceptance interruption. The scan confirms only that the displayed VPA differs from Paytm's registered VPA.

"Where is the Paytm integration?" The signal, the expected identity, the Soundbox context, and the recovery action are all Paytm-native. A standalone app lacks all four.

"How many users does this help?" The official material gives no verified merchant count, so we state the target as Paytm QR and Soundbox merchants and avoid inventing a figure.

"Could you build this today?" Yes. The real core is a typed event replay, a lightweight time-series scorer, a QR decoder, and a VPA comparison. Paytm services are isolated behind an honest mock adapter.

"What if another team copies it?" The title is copyable. The complete mechanism and the physical demo are not. That is why we ship absence modelling, identity verification, device context, and recovery, rather than just an alert.

Final build instruction

Do not begin implementation until the concept is approved. On approval, inspect the existing repository first: framework, entry points, existing components, existing APIs, environment variables, and storage. Do not rewrite working code.

Then implement in vertical slices, each independently demonstrable:

1. Render the seeded normal transaction stream in the existing app shell.
2. Add the real anomaly scorer and the incident alert, with tests.
3. Build the incident explanation and the expected-versus-actual visualisation.
4. Add QR decode and upload, plus deterministic VPA verification.
5. Add the `PaytmAdapter` mock recovery and test-payment confirmation.
6. Harden loading, error, and normal states, mobile responsiveness, mock labels, and demo controls.

Verify each slice before starting the next, so the judge journey stays continuously demonstrable.

The standard

One extremely clear problem. One extremely strong AI insight. One extremely strong Paytm integration. One memorable demo moment. One measurable user outcome.

The product should make judges think: *why doesn't Paytm already have this?* And immediately after: *how did one person build this in a day?*

Primary evidence: the supplied Hyderabad theme PDF and briefing photographs. Prior-field evidence: the supplied 44-entry Mumbai submission CSV. Product verification: public Paytm consumer, business, UPI, and AI Soundbox pages. All quantitative impact beyond stated source facts is labelled as an assumption or a seeded demo value.